

TOI-4087b

R-2141

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_250212 · created 2026-08-12T00:20:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460718.921 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.039
Transit depth (Rp/Rs)^2	1.8 +/- 1.04 [%]
Orbital Inclination (inc)	89.2 +/- 1.99
Airmass coefficient 1 (a1)	0.24 +/- 0.13
Airmass coefficient 2 (a2)	0.85 +/- 0.6
Scatter in the residuals of the lightcurve fit is	57.7 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	6.51
Transit Duration (day)	0.109 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.039 vs 0.10754068997360952 (deviation 0.49σ)
Duration fit vs published	0.109 d vs 0.1242 d (deviation 0.92σ)
Mid-time O–C	5.2 min
Depth SNR	2.5
Residual scatter	57.7 %

Light curve

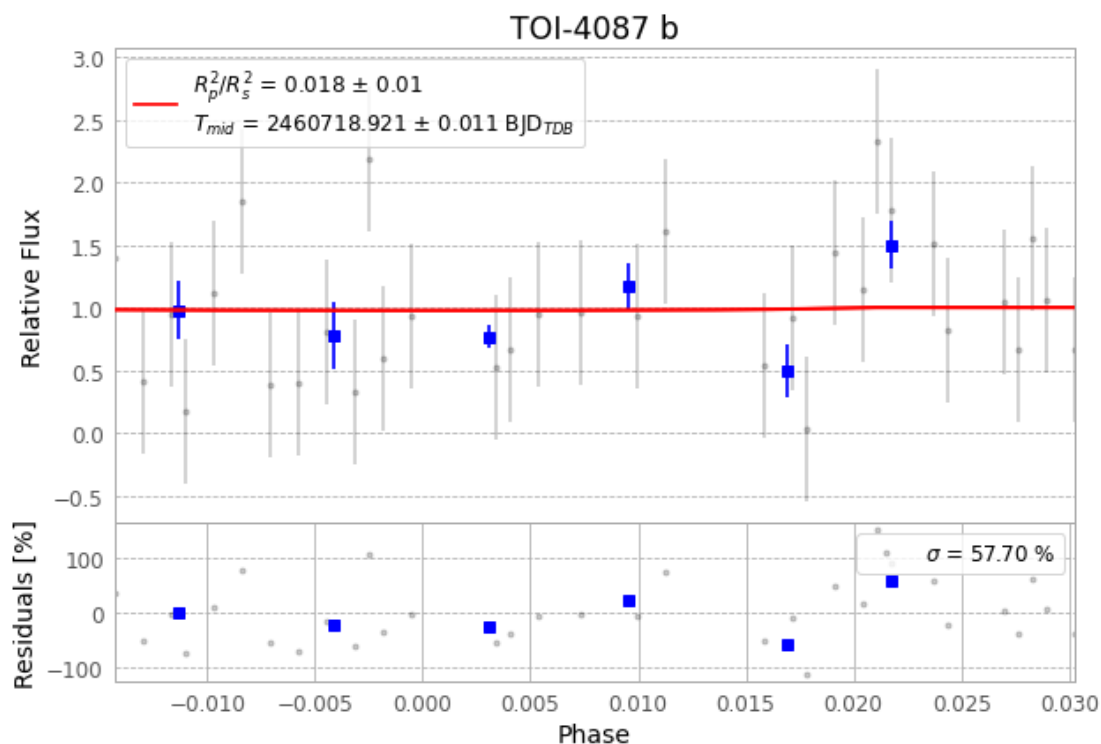


Figure 1 — FinalLightCurve_TOI-4087 b_2025-02-12.png (SHA-256 57eedede2cbd1207...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e5c412cf198d69f6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), TOI4087250212085716.FITS → TOI4087250212122415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-250212.log (183a405cf083...), FinalLightCurve_TOI-4087 b_2025-02-12.png (57eedede2cbd...), FinalLightCurve_TOI-4087 b_2025-02-12.csv (fcf986e8de80...)

Compute resources

Wall time 217.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:20Z.